

Comment 1.4 — Define every symbol immediately below each equation

The manuscript has 11 numbered equations (Sections 3.1.1–3.2, pp. 8–12). Below each equation add a "where ..." line defining **every** symbol at its first use. This is the concrete list — copy the definitions into the paper under each equation.

Eq. (1) — fuzzy membership vector (Sec. 3.1.1)

$$\mu = (\mu_1, \dots, \mu_K), \mu_c \in [0, 1], \sum_c \mu_c = 1, \mu \in \Delta^{K-1}$$

where μ is the fuzzy membership (soft-label) vector, μ_c its component for emotion class c , K the number of emotion classes ($K = 7$), and Δ^{K-1} the $(K-1)$ -dimensional probability simplex.

Eq. (2) — additively-smoothed soft label (Sec. 3.1.1)

$$\mu_c = (n_c + \alpha \cdot \pi_c) / \sum_j (n_j + \alpha \cdot \pi_j)$$

where n_c is the vote count for class c ($n_c = \mathbb{1}[c = y]$ for single-label data), $\alpha > 0$ the smoothing strength, $\pi = (\pi_1, \dots, \pi_K)$ the prior distribution over classes, and j the summation index over the K classes.

Eq. (3) — fuzzy classification loss (Sec. 3.1.1)

$$L_{cls} = - \sum_c \mu_c \log p_c$$

where L_{cls} is the classification loss, $p = (p_1, \dots, p_K)$ the predicted class posterior, p_c its component for class c , and μ_c the fuzzy target of Eq. (2).

Eq. (4) — subspace projection (Sec. 3.1.2)

$$X_{sub} = \text{split}(\text{Conv1}\times\text{1}(X)) = \{X_1, \dots, X_K\}$$

where X is the input feature map, $\text{Conv1}\times\text{1}$ the 1×1 projection to K -d channels, $\text{split}(\cdot)$ the partition into K groups, and X_k the raw sub-feature of emotion k .

Eq. (5) — Gram-Schmidt orthogonalization (Sec. 3.1.2)

$$\tilde{U}_1 = X_1, \tilde{U}_k = X_k - \sum_{\{j < k\}} (\langle X_k, \tilde{U}_j \rangle / \langle \tilde{U}_j, \tilde{U}_j \rangle) \tilde{U}_j$$

where $\tilde{U}_k$ is the k -th orthogonalized sub-feature, $\langle \cdot, \cdot \rangle$ the inner product, k, j subspace indices, and the sum runs over the already-built vectors $j < k$.

Eq. (6) — FLEO block output (Sec. 3.1.2)

$$Y = X + \text{Conv3}\times\text{3}(\parallel_k (\tilde{U}_k \odot s))$$

where $\mathbf{Y}$ is the block output, $\mathbf{X}$ the residual input, **Conv3×3** the fusion convolution, $\|_{\cdot\cdot\mathbf{k}}$ channel-wise concatenation over k , $\tilde{\mathbf{U}}_{\mathbf{k}}$ the orthogonalized sub-feature, $\mathbf{s}$ the SE fuzzy gate, and $\odot$ element-wise (channel) multiplication.

Eq. (7) — total training loss (Sec. 3.1.3)

$$\mathbf{L} = \mathbf{L}_{\text{cls}} + \lambda_o \mathbf{L}_{\text{orth}} + \lambda_b \mathbf{L}_{\text{bind}} = -\sum_c \mu_c \log p_c + \lambda_o \|\hat{\mathbf{U}}\hat{\mathbf{U}}^T - \mathbf{I}\|_{\text{F}}^2 + \lambda_b \text{KL}(\mu \parallel \sigma(\mathbf{b}))$$

where $\mathbf{L}$ is the total loss, λ_o , λ_b the orthogonality and binding weights, $\mathbf{L}_{\text{orth}}$, $\mathbf{L}_{\text{bind}}$ those loss terms, $\hat{\mathbf{U}}$ the stacked orthonormalized bases, $\mathbf{I}$ the identity matrix, $\|\cdot\|_{\text{F}}$ the Frobenius norm, σ the softmax, $\mathbf{b}$ the binding-head logits, and $\text{KL}(\cdot\parallel\cdot)$ the Kullback-Leibler divergence.

Eq. (8) — full inference graph (Sec. 3.2)

$$\mathbf{Y} = \text{Conv3}\times\text{3}(\text{GS}(\text{Conv1}\times\text{1}(\mathbf{X})) \odot \mathbf{s}) + \mathbf{X}$$

where **GS**($\cdot$) is the Gram-Schmidt operator, $\mathbf{s}$ the SE gate, $\odot$ element-wise multiplication, and $\mathbf{X}$, $\mathbf{Y}$, $\text{Conv1}\times\text{1}$, $\text{Conv3}\times\text{3}$ as above.

Eq. (9) — folded (Route-1) inference graph (Sec. 3.2)

$$\mathbf{R} : \mathbf{Y}_{\text{fold}} = \text{Conv3}\times\text{3}(\text{Conv1}\times\text{1}(\mathbf{X}) \odot \mathbf{s}) + \mathbf{X}$$

where $\mathbf{R}$ denotes the fold-out rewrite and $\mathbf{Y}_{\text{fold}}$ the output of the streamlined graph in which GS is replaced by the identity; other symbols as in Eq. (8).

Eq. (10) — fold-out accuracy delta (Sec. 3.2)

$$\Delta_{\text{fold}} = \text{Acc}(\text{full graph, Eq. 8}) - \text{Acc}(\text{folded graph, Eq. 9})$$

where Δ_{fold} is the accuracy drop from folding out GS and **Acc**($\cdot$) the top-1 accuracy of the indicated graph.

Eq. (11) — quantization accuracy delta (Sec. 3.2)

$$\Delta_{\text{q}} = \text{Acc}_{\text{FP32}} - \text{Acc}_{\text{INT8}}$$

where Δ_{q} is the accuracy drop from INT8 quantization, **Acc_{FP32}** the full-precision accuracy and **Acc_{INT8}** the quantized accuracy.

What to do in the paper (Author Action for 1.4)

1. Under **each** of Eq. (1)–(11), insert the corresponding "where ..." sentence above.
2. Make sure every symbol appears with the **same** notation everywhere (e.g. always K for the number of classes, always λ_o / λ_b for the loss weights).

3. Add a one-line **Notation** note or a small symbol table at the start of Section 3 if the reviewer prefers a single reference (optional but clean).